EE 354/CE 361/MATH 310 - Introduction to Probability and Statistics

Fall 2025

EE 354/CE 361/MATH 310 L2 section (Fall 2025)

Introduction to Probability and Statistics -

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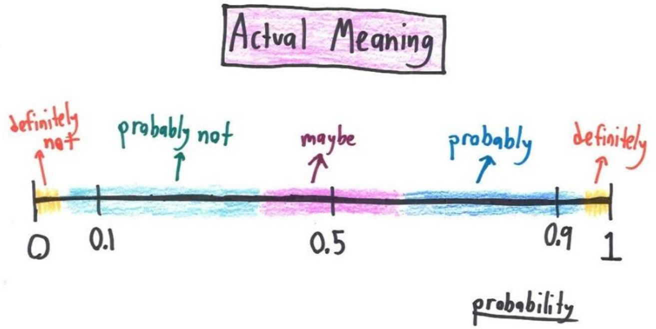
Week 13 - Lecture No 25 – November 12, 2025



Announcements!

1. All lectures and grades posted.
2. *Assignment 3 to be uploaded soon*
3. **No class on Wednesday November 19. 2025 - enjoy**
4. **Quiz No 09 – November 24, 2025 (from Unit 5)**

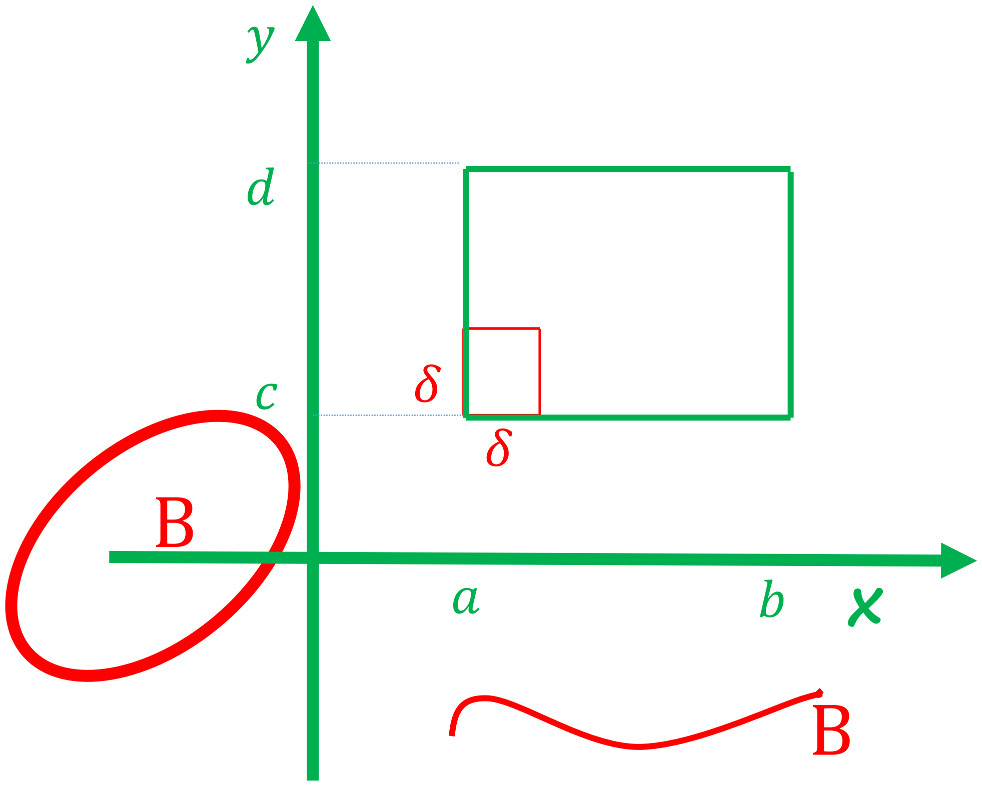
Agenda for today

1. Unit 5 – Continuous Random Variables
   - Joint PDFs
     - **Independent Normal Random Variables**
     - **Bivariate Normal Random Variables**
     - **Uniform joint PDF on a set S**

On Joint PDFs

$$\mathbf{P}\big((X,Y) \in B\big) = \iint_{(X,Y)\in B} f_{X,Y}(x,y)\,dxdy$$

$$\mathbf{P}(\mathrm{a} \le \mathrm{X} \le b, c \le \mathrm{Y} \le d) = \int_c^d \int_a^b f_{X,Y}(x,y)\,dxdy$$

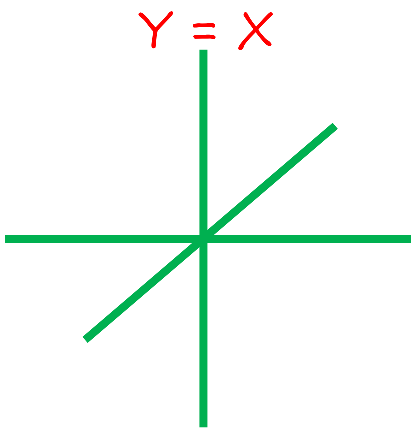


$$\boxed{\mathbf{P}(\mathrm{a} \le \mathrm{X} \le a+\delta, c \le \mathrm{Y} \le c+\delta) \approx f_{X,Y}(\mathrm{a},\mathrm{c})\,\delta^2}$$

Joint PDF   area

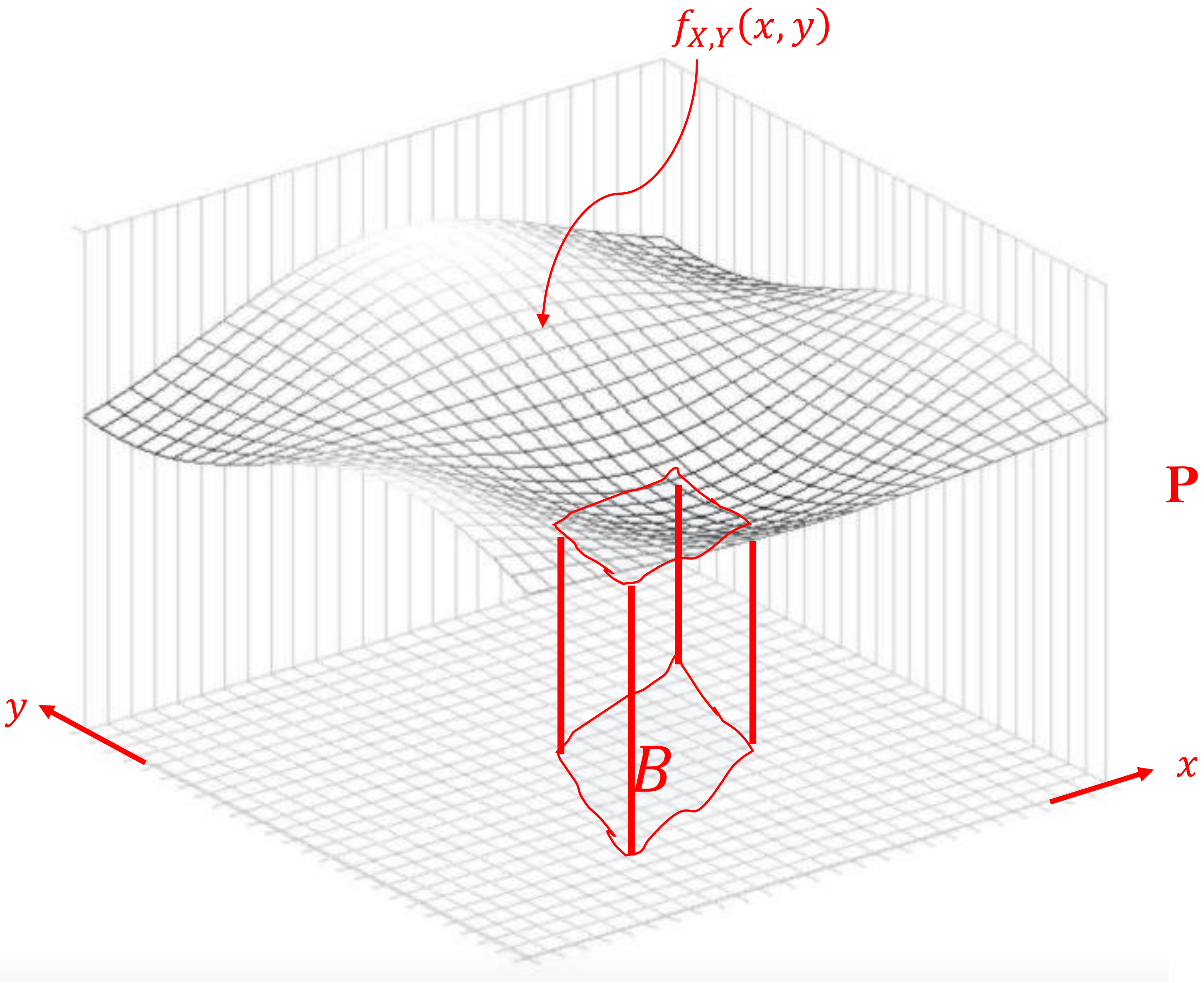
$f_{X,Y}(\mathrm{x,y})$: probability per unit area around the point (x,y)

$area(B) = 0 \implies \mathbf{P}\big((X,Y) \in B\big) = 0$



Are X and Y jointly continuous?

Visualizing a Joint PDF



$$\mathbf{P}\big((X,Y) \in B\big) = \iint_{(X,\,Y)\in B} f_{X,Y}(x,y)\,dxdy$$

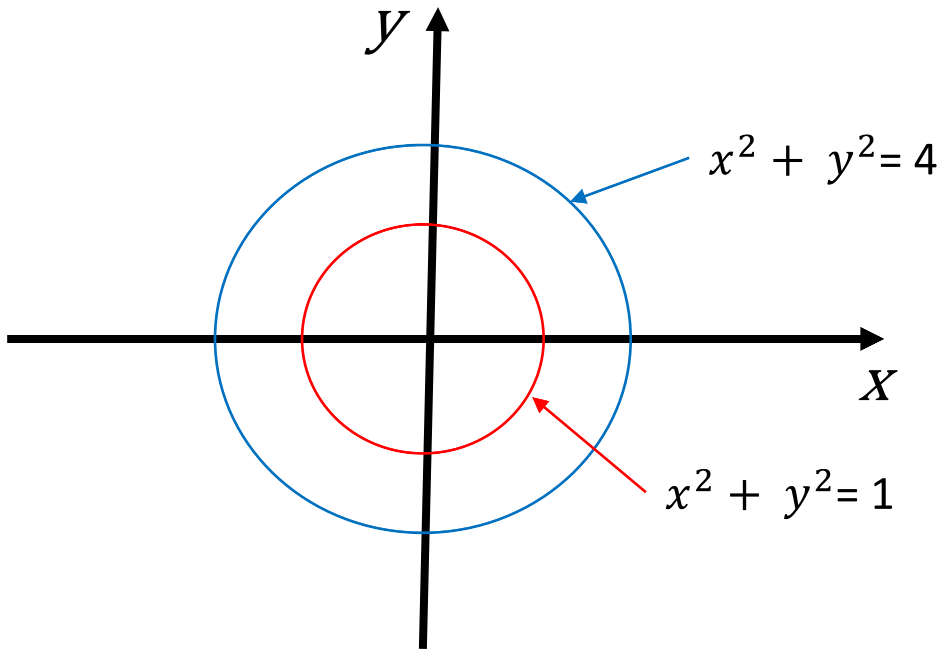
Independent Standard Random Variables

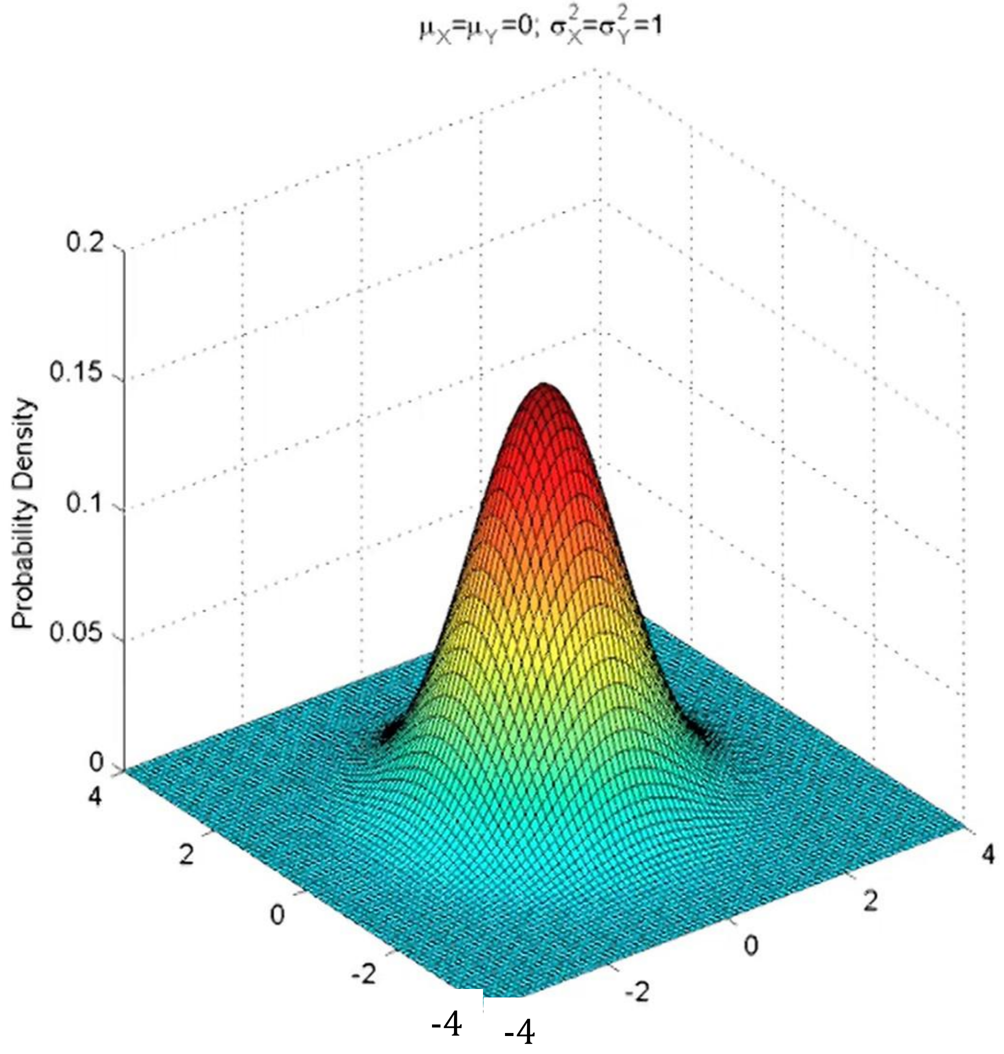
- General normal N(μ,σ^2): $f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

$$f_{X,Y}(x,y) = f_X(x)\,.\,f_Y(y)$$

$$= \frac{1}{\sqrt{2\pi}}\exp\left\{-\frac{x^2}{2}\right\}.\ \frac{1}{\sqrt{2\pi}}\exp\left\{-\frac{y^2}{2}\right\}$$

$$= \frac{1}{2\pi}\exp\left\{-\frac{1}{2}(x^2+\ y^2)\right\}$$

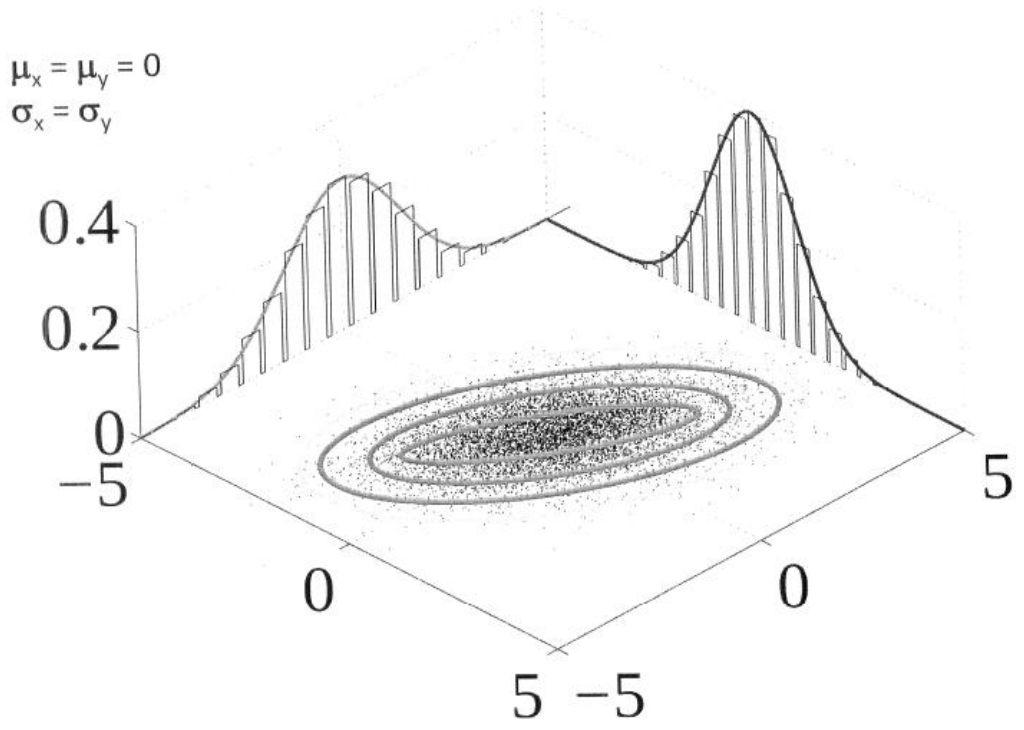


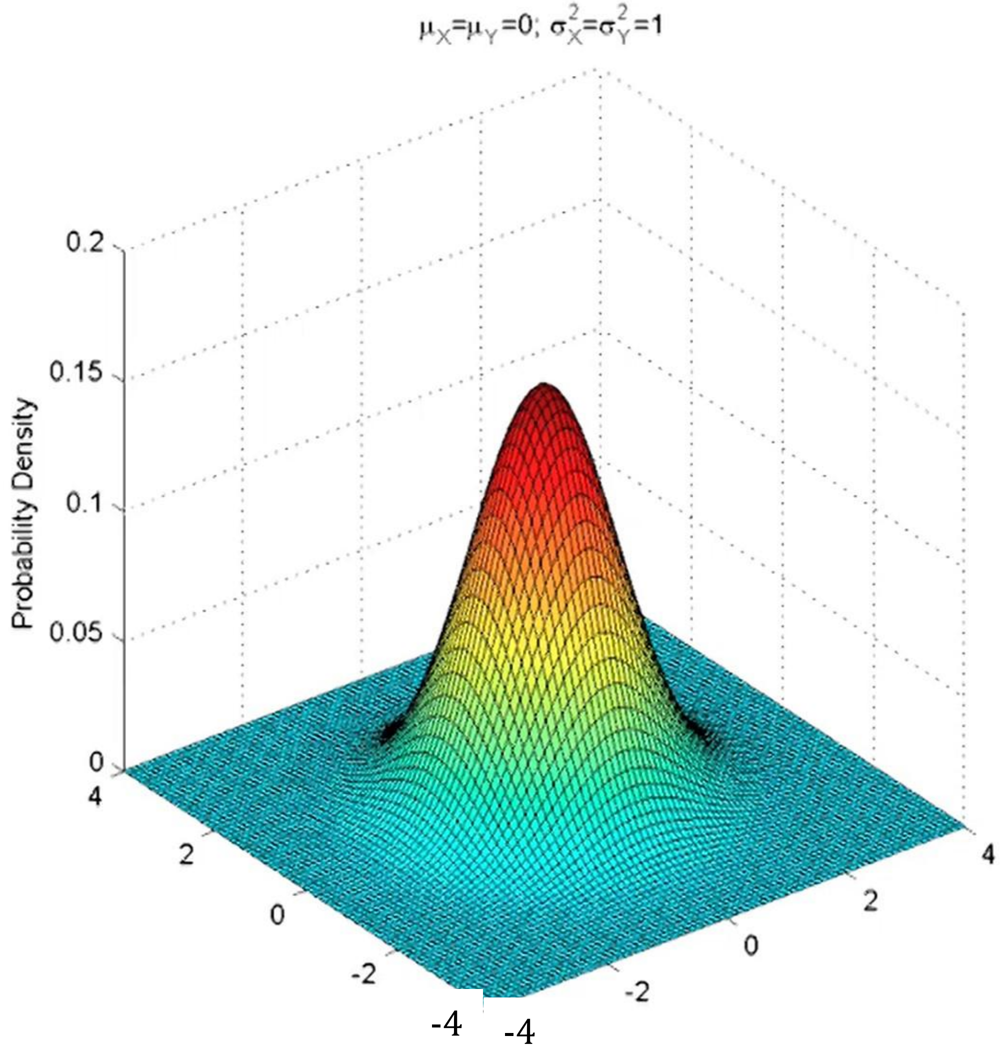


Independent Standard Random Variables

$$f_{X,Y}(x,y) = f_X(x)\,.\,f_Y(y)$$

$$= \frac{1}{\sqrt{2\pi}}\exp\left\{-\frac{x^2}{2}\right\}.\frac{1}{\sqrt{2\pi}}\exp\left\{-\frac{y^2}{2}\right\}$$



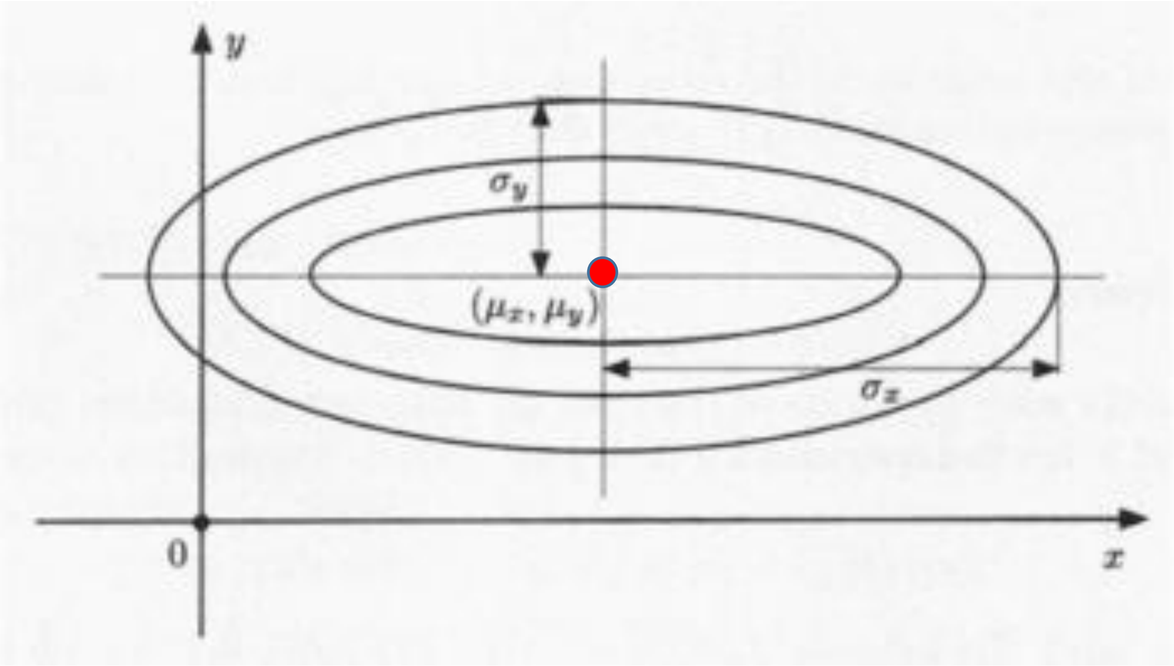


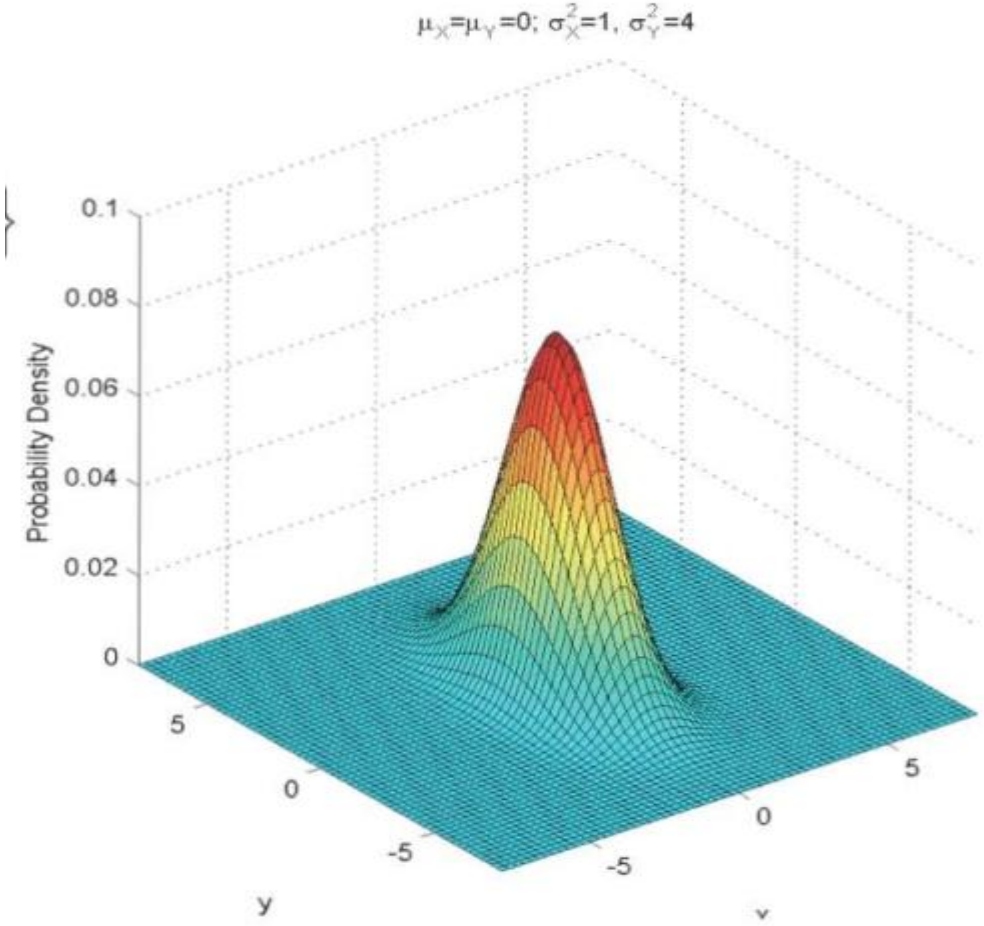
Independent Normal Random Variables

- General normal N(μ,σ^2): $f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

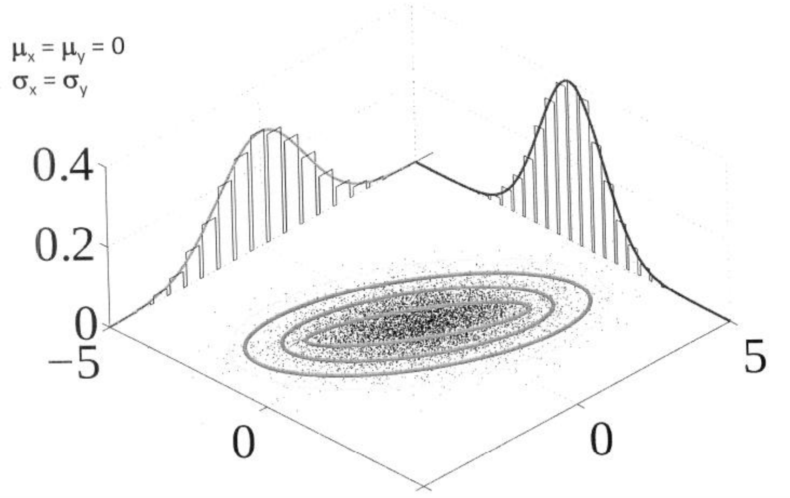
$$f_{X,Y}(x,y) = f_X(x) f_Y(y)$$

$$= \underline{\frac{1}{2\pi\sigma_x\sigma_y}} \exp\left\{-\frac{(x-\mu_x)^2}{2\sigma_x^{\,2}} - \frac{(y-\mu_y)^2}{2\sigma_y^{\,2}}\right\}$$





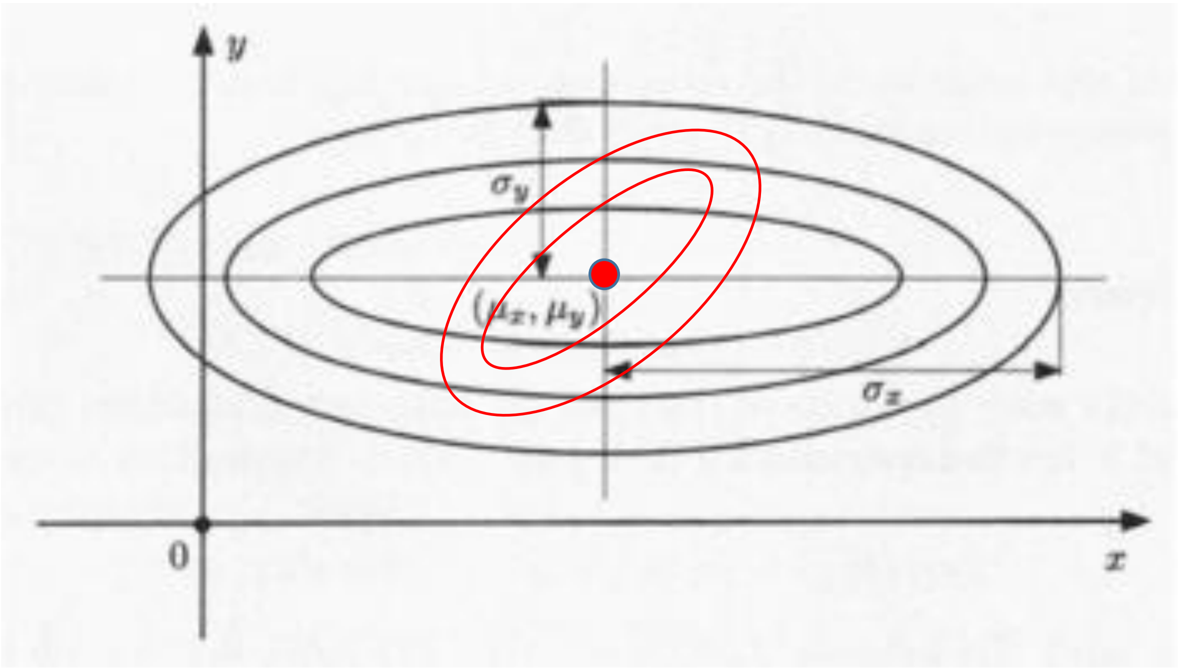
Bivariate Normal Random Variables

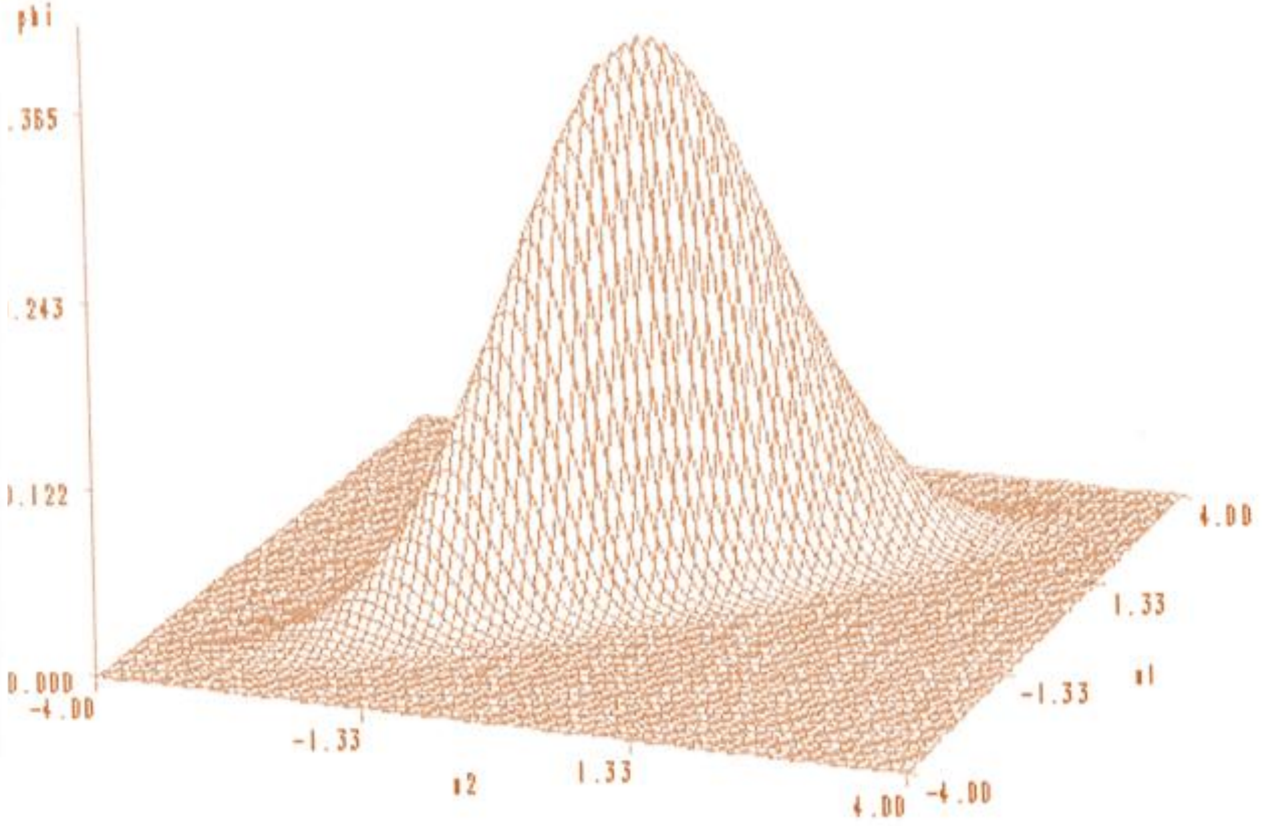


$$f_X(x) = \frac{1}{\sigma_x\sqrt{2\pi}} e^{-\frac{(x-\mu_x)^2}{2\sigma_x{}^2}}$$

$$f_Y(y) = \frac{1}{\sigma_y\sqrt{2\pi}} e^{-\frac{(y-\mu_y)^2}{2\sigma_y{}^2}}$$

$$f_{X,Y}(x, y)$$





From the joint to the marginals

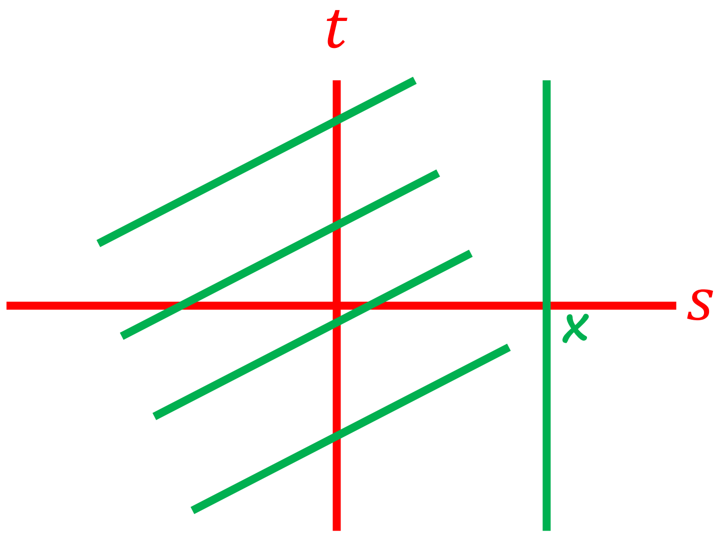
$$p_X(x) = \sum_y P_{X,Y}(x,y)$$

$$f_X(x) = \int f_{X,Y}(x,y)\,dy$$

$$p_Y(y) = \sum_x P_{X,Y}(x,y)$$

$$f_Y(y) = \int f_{X,Y}(x,y)\,dx$$

$$\Rightarrow \quad F_X(x) = P(X \le x) = \int_{-\infty}^{x} \left[\underline{\int_{-\infty}^{\infty} f_{X,Y}(s,t)\,dt} \right] ds$$



$$f_X(x) = \frac{dF_X}{dx}(x) = [\quad]$$

$$= \int_{-\infty}^{\infty} f_{X,Y}(s,t)\,dt$$

Uniform joint PDF on a set S

$$f_{X,Y}(x,y) = \begin{cases} \dfrac{1}{area\ of\ S}, & if\ (x,y)\ \in S, \\ 0, & \text{Otherwise.} \end{cases}$$

$$P(A) = \frac{area\ (A \cap S)}{area\ (S)}$$